

$^{36}\text{Ar}(\text{n},\text{d})$ 1977Pa29

$J^\pi=0^+$ for ^{36}Ar ground state.

1977Pa29: 14.1-MeV neutrons were produced via the $^3\text{H}(\text{d},\text{n})^4\text{He}$ reaction at an average rate of 10^9 neutrons per second into 4π sr from the Brown University neutron generator and impinged on a 2.85-mg/cm^2 ^{36}Ar target in the form of a gas cell filled with 99.8% enriched ^{36}Ar . Deuterons were detected using a time-of-flight counter telescope consisting of two proportional counters and a NaI(Tl) crystal. Measured $\sigma(E_{\text{d}},\theta)$. Deduced levels, L, spectroscopic factors from local zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

 ^{35}Cl Levels

<u>E(level)[†]</u>	<u>L[‡]</u>	<u>Comments</u>
0	2	S: 3.83 48 and 3.90 47 using two sets of optical parameters.
1219 [#]	0 [#]	S: 1.94 23 and 3.30 40 using two sets of optical parameters. Other spectroscopic factors calculated using two sets of optical parameters did not show large differences.
1763 [#]	(2) [#]	

[†] From 1977Pa29.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in 1977Pa29.

[#] 1977Pa29 state that the first and second excited states are recorded together and not resolved, but their angular distributions could be separated due to different L-transfers, however statistical errors prevent an accurate determination of the smaller L=2 component from the second excited state.